

## EMEI SHAN, China

WMO: 563850

Lat: 29.5214N

Lon: 103.3356E

Elev: 3049

StdP: 69.67

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|------|-------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -13.2      | -11.8 | -18.6                         | 1.1 | 0.1  | -16.0 | 1.3 | -2.6 | 13.2                  | 2.0  | 11.8 | 1.2  | 2.1                   | 320  | 0.702 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 4.3                    | 17.6            | 13.0 | 16.5 | 12.9 | 15.5 | 12.6 | 15.1                | 16.3 | 14.5 | 15.5 | 13.9 | 14.7 | 2.7                  | 250  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 14.7                            | 15.3 | 15.7 | 14.1 | 14.7 | 15.1 | 13.5 | 14.2 | 14.5 | 54.9          | 16.5 | 52.7 | 15.6 | 50.5 | 14.8 | 21.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |     |                    |      | n-Year Return Period Values of Extreme Temperature |     |            |      |            |      |            |      |       |      |
|-------------------|------|-----|----------------------------|-----|--------------------|------|----------------------------------------------------|-----|------------|------|------------|------|------------|------|-------|------|
|                   |      |     | Mean                       |     | Standard Deviation |      | n=5 years                                          |     | n=10 years |      | n=20 years |      | n=50 years |      |       |      |
| 1%                | 2.5% | 5%  | Min                        | Max | Min                | Max  | Min                                                | Max | Min        | Max  | Min        | Max  | Min        | Max  |       |      |
| (a)               | (b)  | (c) | (e)                        | (f) | (g)                | (h)  | (i)                                                | (j) | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |       |      |
| (4)               | 10.1 | 8.5 | 7.3                        | DB  | -14.7              | 20.8 | 1.8                                                | 1.8 | -16.0      | 22.0 | -17.1      | 23.1 | -18.1      | 24.1 | -19.3 | 25.4 |
| (5)               |      |     |                            | WB  | -15.2              | 16.7 | 1.7                                                | 1.0 | -16.5      | 17.5 | -17.5      | 18.1 | -18.5      | 18.6 | -19.7 | 19.4 |

## Monthly Climatic Design Conditions

|      |                                                                   | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |       |      |
|------|-------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|      |                                                                   | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |       |      |
| (6)  | Temperatures, Degree-Days and Degree-Hours                        | DBAvg       | 4.2   | -5.0  | -3.0  | 0.1   | 4.4   | 7.3   | 10.2  | 12.5  | 11.9  | 9.0   | 4.7   | 1.1   | -3.3  | (6)  |
| (7)  |                                                                   | DBStd       | 6.95  | 4.74  | 5.16  | 4.60  | 4.23  | 3.00  | 2.46  | 1.85  | 2.14  | 2.89  | 3.26  | 4.00  | 4.08  | (7)  |
| (8)  |                                                                   | HDD10.0     | 2343  | 464   | 364   | 309   | 174   | 92    | 27    | 3     | 8     | 52    | 168   | 269   | 413   | (8)  |
| (9)  |                                                                   | HDD18.3     | 5161  | 722   | 597   | 566   | 419   | 341   | 245   | 180   | 199   | 279   | 424   | 518   | 672   | (9)  |
| (10) |                                                                   | CDD10.0     | 223   | 0     | 0     | 2     | 5     | 9     | 32    | 81    | 67    | 23    | 2     | 1     | 0     | (10) |
| (11) |                                                                   | CDD18.3     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | (11) |
| (12) |                                                                   | CDH23.3     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | (12) |
| (13) |                                                                   | CDH26.7     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | (13) |
| (14) | Wind                                                              | WSAvg       | 2.9   | 3.1   | 3.2   | 3.0   | 3.0   | 2.8   | 2.7   | 2.6   | 2.6   | 2.7   | 2.5   | 2.9   | 3.0   | (14) |
| (15) | Precipitation                                                     | PrecAvg     | 1863  | 16    | 24    | 53    | 114   | 167   | 231   | 391   | 475   | 216   | 111   | 47    | 17    | (15) |
| (16) |                                                                   | PrecMax     | 2506  | 38    | 49    | 92    | 189   | 323   | 446   | 739   | 912   | 421   | 223   | 101   | 28    | (16) |
| (17) |                                                                   | PrecMin     | 1304  | 6     | 8     | 24    | 53    | 110   | 122   | 157   | 254   | 84    | 47    | 22    | 5     | (17) |
| (18) |                                                                   | PrecStd     | 296   | 7     | 10    | 20    | 37    | 49    | 90    | 154   | 173   | 93    | 44    | 17    | 7     | (18) |
| (19) | Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB    | 11.9  | 13.8  | 15.6  | 17.4  | 17.7  | 18.1  | 18.6  | 18.6  | 17.6  | 15.0  | 13.8  | 11.7  | (19) |
| (20) |                                                                   | MCWB        | 1.3   | 2.1   | 4.7   | 7.8   | 10.0  | 13.0  | 14.2  | 14.8  | 13.5  | 10.5  | 4.4   | 1.4   | 1.4   | (20) |
| (21) |                                                                   | 2%          | DB    | 8.4   | 10.7  | 12.2  | 14.5  | 14.8  | 16.2  | 17.2  | 17.0  | 15.7  | 12.3  | 10.5  | 8.0   | (21) |
| (22) |                                                                   | MCWB        | -0.3  | 1.4   | 3.9   | 7.4   | 9.9   | 12.5  | 14.0  | 14.3  | 13.0  | 8.8   | 4.1   | 0.2   | 0.2   | (22) |
| (23) |                                                                   | 5%          | DB    | 5.5   | 7.8   | 9.6   | 12.0  | 12.9  | 14.8  | 16.2  | 15.9  | 14.1  | 10.5  | 8.2   | 5.2   | (23) |
| (24) |                                                                   | MCWB        | -1.1  | 0.2   | 3.2   | 6.9   | 9.3   | 12.2  | 13.7  | 13.7  | 12.2  | 8.3   | 3.8   | -0.5  | -0.5  | (24) |
| (25) |                                                                   | 10%         | DB    | 2.8   | 5.2   | 6.8   | 9.9   | 11.5  | 13.6  | 15.2  | 14.8  | 12.8  | 8.9   | 6.1   | 2.8   | (25) |
| (26) |                                                                   | MCWB        | -2.0  | -0.7  | 2.4   | 6.4   | 8.9   | 11.6  | 13.3  | 13.2  | 11.4  | 7.5   | 3.1   | -1.5  | -1.5  | (26) |
| (27) | Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB    | 1.8   | 3.1   | 7.0   | 10.7  | 12.7  | 15.4  | 16.0  | 16.0  | 14.8  | 11.7  | 6.9   | 2.5   | (27) |
| (28) |                                                                   | MCDWB       | 10.4  | 11.6  | 11.4  | 13.6  | 14.3  | 16.7  | 17.3  | 17.7  | 16.3  | 13.7  | 10.4  | 8.7   | 8.7   | (28) |
| (29) |                                                                   | 2%          | WB    | 0.2   | 1.9   | 5.1   | 9.2   | 11.5  | 14.1  | 15.2  | 15.0  | 13.7  | 10.1  | 5.6   | 1.1   | (29) |
| (30) |                                                                   | MCDWB       | 6.8   | 9.1   | 9.4   | 11.6  | 12.9  | 15.0  | 16.3  | 16.1  | 15.0  | 11.1  | 8.4   | 6.3   | 6.3   | (30) |
| (31) |                                                                   | 5%          | WB    | -0.8  | 0.7   | 4.0   | 8.1   | 10.5  | 13.1  | 14.6  | 14.3  | 12.8  | 8.8   | 4.6   | 0.0   | (31) |
| (32) |                                                                   | MCDWB       | 4.8   | 6.7   | 7.6   | 10.5  | 12.0  | 13.9  | 15.5  | 15.3  | 13.7  | 9.8   | 6.8   | 3.9   | 3.9   | (32) |
| (33) |                                                                   | 10%         | WB    | -1.9  | -0.4  | 2.9   | 7.0   | 9.5   | 12.2  | 14.0  | 13.7  | 11.8  | 7.6   | 3.6   | -1.0  | (33) |
| (34) |                                                                   | MCDWB       | 2.4   | 4.1   | 6.1   | 9.1   | 10.9  | 13.0  | 14.7  | 14.5  | 12.5  | 8.6   | 5.3   | 2.1   | 2.1   | (34) |
| (35) | Mean Daily Temperature Range                                      |             | MDBR  | 7.0   | 7.1   | 6.5   | 6.2   | 5.4   | 4.6   | 4.3   | 4.4   | 4.5   | 5.8   | 6.3   | 6.3   | (35) |
| (36) |                                                                   | 5% DB       | MCDBR | 11.9  | 12.0  | 11.3  | 9.8   | 7.7   | 6.4   | 5.7   | 5.7   | 6.0   | 7.0   | 9.3   | 10.8  | (36) |
| (37) |                                                                   |             | MCWBR | 6.0   | 5.7   | 5.9   | 5.9   | 6.0   | 5.3   | 4.8   | 4.6   | 4.4   | 4.9   | 4.7   | 5.4   | (37) |
| (38) |                                                                   |             | MCDBR | 10.9  | 10.9  | 9.0   | 7.7   | 6.5   | 5.2   | 4.8   | 5.2   | 5.3   | 6.0   | 7.5   | 9.2   | (38) |
| (39) |                                                                   | 5% WB       | MCWBR | 6.2   | 5.9   | 5.6   | 5.4   | 5.2   | 4.4   | 4.0   | 4.2   | 3.8   | 4.4   | 4.5   | 5.4   | (39) |
| (40) | Clear-Sky Solar Irradiance                                        | taub        | 0.366 | 0.382 | 0.410 | 0.405 | 0.388 | 0.375 | 0.395 | 0.406 | 0.406 | 0.399 | 0.391 | 0.371 | 0.371 | (40) |
| (41) |                                                                   | taud        | 2.100 | 2.094 | 2.060 | 2.142 | 2.249 | 2.328 | 2.266 | 2.218 | 2.190 | 2.136 | 2.074 | 2.089 | 2.089 | (41) |
| (42) |                                                                   | Ebn at Noon | 848   | 873   | 873   | 890   | 902   | 909   | 889   | 877   | 866   | 843   | 812   | 818   | 818   | (42) |
| (43) |                                                                   | Edh at Noon | 133   | 147   | 162   | 155   | 140   | 129   | 136   | 142   | 141   | 139   | 136   | 129   | 129   | (43) |
| (44) | All-Sky Solar Radiation                                           | RadAvg      | 1.92  | 2.42  | 3.05  | 3.79  | 3.96  | 3.69  | 4.12  | 3.95  | 2.79  | 2.17  | 1.98  | 1.74  | 1.74  | (44) |
| (45) |                                                                   | RadStd      | 0.30  | 0.34  | 0.35  | 0.42  | 0.50  | 0.53  | 0.53  | 0.60  | 0.34  | 0.31  | 0.15  | 0.27  | 0.27  | (45) |

## Historical Trends

|      |                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|      |                        | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) | Station Only           | N/A   | N/A     | N/A    | +0.42   | +0.27 | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (7 neighbors) | +0.31 | N/A     | -0.60  | +0.49   | N/A   | N/A   | N/A         | -72     | +148    | +81     |

Nomenclature: See separate page